THE LENGTH OF A QUASI-COHERENT MODULE AT A POINT IS UNCHANGED BY PUSHFORWARD ALONG A CLOSED IMMERSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: an etale chart reduction to closed immersions of schemes, an identification of the restricted pushforward with the pushforward of the restricted module, and a length comparison through the surjection of local rings

Editorial note. The label lemma-length-closed-immersion names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The length of a quasi-coherent module at a point, and its compatibility with etale charts
- Etale charts and the cartesian description of the points of a fibre product, together with stability of closed immersions under base change
- The identification of the pushforward along a base changed closed immersion with the restriction of the pushforward, and the algebra statement that the length of a module over a local ring agrees with its length viewed through a surjection of local rings

Source reference. Original lemma 00IX: sources/stacks-chow/tags.txt, source lines 695–695. Prerequisite pointer invoked at proof line(s) 555; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original equation 071Z: sources/stacks-chow/tags.txt, source lines 9008–9008. Prerequisite pointer invoked at proof line(s) 545; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02YW: sources/stacks-chow/tags.txt, source lines 3891–3891. Prerequisite pointer invoked at proof line(s) 540; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03H4: sources/stacks-chow/tags.txt, source lines 4553–4553. Prerequisite pointer invoked at proof line(s) 539; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let $i : Y \to X$ be a closed immersion of algebraic spaces over S. Let $\mathcal{G}$ be a quasi-coherent $\mathcal{O}_Y$-module. Let $y \in |Y|$ with image $x \in |X|$. Let $d \in \{0, 1, 2, \ldots, \infty\}$. The following are equivalent*

(1) *$\mathcal{G}$ has length d at y, and*
(2) *$i_*\mathcal{G}$ has length d at x.*

Proof. Choose an étale morphism $f : U \to X$ with U a scheme and $u \in U$ mapping to x. Set $V = Y \times_X U$. Denote $g : V \to Y$ and $j : V \to U$ the projections. Then $j : V \to U$ is a closed immersion and there is a unique point $v \in V$ mapping to $y \in |Y|$ and $u \in U$ (use Properties of Spaces, Lemma 03H4 and Spaces, Lemma 02YW). We have $j_*(\mathcal{G}|_V) = (i_*\mathcal{G})|_U$ as modules on the scheme V and j_* the "usual" pushforward of modules for the morphism of schemes j, see discussion surrounding Cohomology of Spaces, Equation (071Z). In this way we reduce to the case of schemes: if $i : Y \to X$ is a closed immersion of schemes, then

$$(i_*\mathcal{G})_x = \mathcal{G}_y$$

as modules over $\mathcal{O}_{X,x}$ where the module structure on the right hand side is given by the surjection $i_y^\sharp : \mathcal{O}_{X,x} \to \mathcal{O}_{Y,y}$. Thus equality by Algebra, Lemma 00IX. □